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IMAGE FORMATION APPARATUS, METHOD FOR THE IMAGE FORMATION APPARATUS, AND RECORDING MEDIUM

Abstract

Provided is an image formation apparatus including a reading unit configured to read a sheet on which a character string is printed, and a memory storing a program and a processor, when executing the program, configured to extract the character string from the read sheet, generate, based on the extracted character string, login information for logging into the image formation apparatus, and register the generated login information.

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to an image formation apparatus, a method for the image formation apparatus, and a recording medium.

Description of the Related Art

[0002] Personal information of a user who uses an image formation apparatus is registered by operating a touch panel of the image formation apparatus. When the image formation apparatus is to be used, an authentication is performed to use a function of the image formation apparatus. As a user registration method in a multi-functional peripheral, Japanese Patent Laid-Open No. 2005-335215 describes registration of a user in an address book by the user from an operation panel of the multi-functional peripheral.

[0003] According to the above-described technique, registration of user information that is necessary to use the apparatus needs to be performed by the user via an operation unit. The registration of the user information by the user involves large operational burden, and an operational error may occur.

SUMMARY

[0004] An image formation apparatus according to an aspect of the present disclosure includes a reading unit configured to read a sheet on which a character string is printed, and a memory storing a program and a processor, when executing the program, configured to extract the character string from the read sheet, generate, based on the extracted character string, login information for logging into the image formation apparatus, and register the generated login information.

[0005] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates a configuration of an entire printing system according to a first embodiment of the present disclosure.

[0007] FIG. 2 illustrates a hardware configuration of a client terminal according to the first embodiment of the present disclosure.

[0008] FIG. 3 illustrates a hardware configuration of an image formation apparatus according to the first embodiment of the present disclosure.

[0009] FIG. 4 illustrates a software configuration of the entire printing system according to the first embodiment of the present disclosure.

[0010] FIG. 5 illustrates a flowchart of user registration processing of the image formation apparatus according to the first embodiment of the present disclosure.

[0011] FIG. 6 illustrates descriptions of an identification certificate according to the first embodiment of the present disclosure.

[0012] FIG. 7 illustrates a software configuration of the entire printing system according to a second embodiment of the present disclosure.

[0013] FIG. 8 illustrates descriptions of a registration job according to the second embodiment of the present disclosure.

[0014] FIG. 9 illustrates a sequence of the entire printing system according to the second embodiment of the present disclosure.

[0015] FIG. 10 illustrates a flowchart of provisional registration processing of the image formation apparatus according to a third embodiment of the present disclosure.

[0016] FIG. 11 illustrates a flowchart of formal registration processing of the image formation apparatus according to the third embodiment of the present disclosure.

[0017] FIG. **12** illustrates a flowchart of user information update processing of the image formation apparatus according to a fourth embodiment of the present disclosure.

[0018] FIG. **13** illustrates descriptions of the registration job according to a fifth embodiment of the present disclosure.

[0019] FIG. **14** illustrates a sequence of the entire printing system according to the fifth embodiment of the present disclosure.

[0020] FIG. **15** illustrates a flowchart of user information registration processing of the image formation apparatus which has received user information according to the fifth embodiment of the present disclosure.

[0021] FIG. **16** illustrates a flowchart of the user information registration processing of the image formation apparatus according to a sixth embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0022] Each embodiment of the present disclosure will be described with reference to the attached drawings. The following embodiments are not intended to be limiting, and not all combinations of features described in the respective embodiments are necessarily essential.

First Embodiment

[0023] FIG. **1** is a system configuration diagram illustrating an entire printing system according to a first embodiment of the present disclosure.

[0024] A client terminal **101** has a function of issuing an execution instruction such as printing or scanning to an image formation apparatus **102** by using a browser or the like. A personal computer or a mobile terminal is assumed as the client terminal **101**, but the type of the client terminal **101** is not limited.

[0025] The image formation apparatus **102** has a scanning function with which image data can be obtained from a recording medium placed on the image formation apparatus **102**. The image formation apparatus **102** also has a user management function of authenticating a user of the image formation apparatus **102** to impose a use restriction of the function such as printing or scanning.

[0026] One or a plurality of client terminals **101** and one or a plurality of image formation apparatuses **102** are connected to each other via a network **100**.

[0027] FIG. **2** illustrates a hardware configuration of the client terminal **101** according to the first embodiment of the present disclosure.

[0028] A central processing unit (CPU) **200** controls the client terminal **101**. The CPU **200** executes a program stored in a ROM **202** or a secondary storage device **205** and temporarily stores information used to execute the program in a RAM **201**. The random access memory (RAM) **201** is a storage unit and temporarily stores information used to execute the program. The read only memory (ROM) **202** is a storage unit and stores various data such as a basic I/O program therein.

[0029] A network I/F (interface) **203** enables the client terminal **101** to communicate with the other client terminal **101** or the image formation apparatus **102**. A communication method supports both wired communication and wireless communication. A system bus **204** connects each element in the client terminal **101** and propagates a control command from the CPU **200** across the client terminal **101**. The secondary storage device **205** is an external storage unit and functions as a large volume memory. The secondary storage device **205** includes a hard disk drive (HDD), a flash memory, or the like and stores a program or the like of a web browser or a service server group.

[0030] A keyboard **206** is an instruction input unit. A display **207** is a display unit, and when the display **207** is connected to the image formation apparatus **102**, setting information associated with the image formation apparatus **102** can be browsed. An interface **208** is an external apparatus I/F and enables connection to the image formation apparatus **102** or a peripheral such as a USB device.

[0031] The above-described configuration is an example of the present embodiment and is not limited to the configuration example illustrated in FIG. **2**. For example, the RAM **201** can be designated as a storage destination of the program, and the program can be executed by the CPU **200**.

[0032] FIG. 3 illustrates a hardware configuration of the image formation apparatus **102** according to the first embodiment of the present disclosure.

[0033] A central processing unit (CPU) **300** controls the image formation apparatus **102**. The CPU **300** executes a program stored in a ROM **302** or a secondary storage device **308** and temporarily stores information used to execute the program in the RAM **301**. The random access memory (RAM) **301** is a storage unit and temporarily stores information used to execute the program. The read only memory (ROM) **302** is a storage unit and stores various data such as a basic I/O program therein.

[0034] A network I/F **303** enables the image formation apparatus **102** to communicate with the client terminal **101** or the other image formation apparatus **102**. A communication method supports both wired communication and wireless communication.

[0035] A system bus **304** connects each element in the image formation apparatus **102** and propagates a control command from the CPU **300** across the image formation apparatus **102**. A device controller **305** controls a device such as a printer **306** or a scanner **307** based on the control command received from the CPU **300**.

[0036] The printer **306** executes a printing job. The scanner **307** executes scanning of the recording medium placed on the image formation apparatus **102**. A hard disc drive (HDD) or a flash memory is applicable as the secondary storage device **308**. With an input/output I/F **309**, a signal received from an input/output apparatus **310** is transferred to each element in the image formation apparatus **102**.

[0037] The input/output apparatus **310** represents a plurality of configurations that perform input and output operations in the image formation apparatus **102**. Specifically, the input/output apparatus **310** includes a touch panel, a hard key, and the like, and accepts an input from the user to transmit a signal corresponding to the input to each element in the image formation apparatus **102** via the input/output I/F **309**.

[0038] FIG. 4 illustrates a software configuration according to the present embodiment. A driver **400** is a function included in the client terminal **101**. The driver **400** generates a job for controlling the image formation apparatus **102** and transmits the job to the image formation apparatus **102** via the network I/F **203** to enable operation of the image formation apparatus **102**.

[0039] A scanning unit **401** scans an identification certificate placed on the image formation apparatus **102** to obtain image data of the identification certificate. The scanning unit **401** executes scanning when the scanner **307** operates based on the job transmitted from the driver **400** or the signal transmitted from the input/output apparatus **310**.

[0040] An extraction processing unit **402** subjects the image data obtained by the scanning unit **401** to optical character reader (OCR) processing to obtain text information included in the image data. The extraction processing unit **402** then obtains a predetermined character string from the text information. Specifically, the predetermined character string includes a name, a personal number, date information indicating an expiry, or the like. The processing of obtaining the text information from the image data may be a processing other than the OCR processing.

[0041] A user information generation unit **403** generates user information used to log into the image formation apparatus **102** using predetermined character strings obtained by the extraction processing unit **402**. Specifically, the user information generation unit **403** selects one of the predetermined character strings as a user ID and another character string as an initial password to generate user information in which a combination of the user ID and the initial password is stored. The user information generation unit **403** also attaches a hash value to the user information to uniquely identify the user information. When the generation of the user information is completed, the user information generation unit **403** stores the generated user information in the secondary storage device **308**.

[0042] An example of the user information (login information) generated by the user information generation unit **403** is illustrated in Table 1.

TABLE-US-00001 TABLE 1 User information User ID Password HashKey Shimomaru
Taro 110100197810270022 12345abcdef

[0043] A “user ID” column stores a value for identifying a user of the image formation apparatus **102**. In the case of this example, the user information generation unit **403** selects, as a first character string, information relevant to the name from the text information obtained by the extraction processing unit **402** to be set as a value in the column of the “user ID”.

[0044] A “password” column stores an initial value of the password that is used by a person whose data is printed on the identification certificate to log into the image formation apparatus **102**. In the case of this example, the user information generation unit **403** selects, as a second character string, information relevant to the personal number with which the person whose data is printed on the identification certificate can be uniquely identified from the text information obtained by the extraction processing unit **402** to be set as a value in the “password” column. It is noted that the value in the “password” column is not limited to the personal number described on the identification certificate. For example, other information described on the identification certificate such as a date of expiry of the identification certificate may be used.

[0045] A “HashKey” column stores a hash value for uniquely identifying the user information.

[0046] The user of the image formation apparatus **102** can log into the image formation apparatus **102** by inputting the values in the “user ID” column and the “initial password” column from the input/output apparatus **310**.

[0047] An external communication unit **404** communicates with the client terminal **101** to receive the job generated by the driver **400**. Thus, the image formation apparatus **102** causes the scanning unit **401**, a printing unit **405**, a copying unit **406**, and a boxing unit **407** to operate according to content of the received job to provide each function of the image formation apparatus **102** to the user.

[0048] A user information management unit **408** refers to the secondary storage device **308** to manage all the user information in Table 1 as a “user table”. The user information management unit **408** performs login processing to the image formation apparatus **102**, display of the user information described in Table 1 via the input/output apparatus **310**, and set value alteration processing by using the user table. An example of the user table is illustrated in Table 2.

TABLE-US-00002 TABLE 2 User table User ID Password HashKey Shimomaru
Taro 110100197810270022 12345abcdef Shimomaru
Hanako 110100197810270023 67890uvwxyz

[0049] Each column in the user table is equivalent to each column of the user information in Table 1, and the description thereof is not provided herein.

[0050] FIG. 5 illustrates a flowchart of user registration processing of the image formation apparatus **102** according to the present embodiment. The flowchart in FIG. 5 is realized when the CPU **300** of the image formation apparatus **102** reads out a program for realizing the features of the flowchart in FIG. 5 stored in the ROM **302** or the secondary storage device **308** onto the RAM **301** to execute the program. A process flow in FIG. 5 is started when the scanning unit **401** is activated by a signal originating from the input/output apparatus **310** or a scanning instruction transmitted from the driver **400**.

[0051] In S500, the scanning unit **401** scans (reads) a sheet (for example, an identification certificate) on which a character string is printed and which is placed on the scanner **307** to obtain image data. In S501, text information is extracted from the image data obtained in S500 by the extraction processing unit **402**. The text information to be extracted will be described with reference to FIG. 6.

[0052] FIG. 6 illustrates an example of the identification certificate to be scanned by the image formation apparatus **102** according to the present embodiment.

[0053] Name **600** indicates a name of an owner of the identification certificate. Gender **601** indicates a gender of the owner of the identification certificate, including an option other than male or female (for example, other). Date of birth **602** indicates a date of birth of the owner of the

identification certificate. Address **603** indicates an address of the owner of the identification certificate. Expiry **604** is date information indicating an expiry of the identification certificate. A format of the date information is not limited to the example illustrated in FIG. 6. For example, the format can also include a format “YYYYMMDDThhmmss” defined in accordance with ISO 8601. [0054] Issuing source **605** indicates an issuer of the identification certificate. National identification number **606** is a number uniquely allocated to the owner of the identification certificate. The information is not limited to the national identification number as long as the information is described on the identification certificate subjected to scanning and is uniquely allocated to the owner of the identification certificate. For example, an employee number may be used in the case of an employee identification certificate indicating affiliation to a company.

[0055] A language display of the information from the name **600** to the national identification number **606** is not limited.

[0056] The description will now return to FIG. 5.

[0057] It is assumed that the text information to be extracted in **S501** includes all pieces of character information described on the identification certificate from the name **600** to the national identification number **606** in FIG. 6.

[0058] In **S502**, it is determined whether the text information has been successfully extracted by the extraction processing unit **402** in **S501**. When it is determined that the text information has been successfully extracted, in other words, when the information of at least one of the name **600** to the national identification number **606** has been successfully extracted, the process flow transitions to step **S503**. When the text information has not been successfully extracted, in other words, none of the information from the name **600** to the national identification number **606** has been extracted, the process flow in FIG. 5 ends.

[0059] In **S503**, a character string to be used to generate the user information described in Table 1 is extracted from the text information generated in **S501** by the extraction processing unit **402**. Specifically, the extraction processing unit **402** extracts, from the text information, the information relevant to the name **600** as the first character string and the information relevant to the national identification number **606** or the expiry **604** as the second character string.

[0060] In **S504**, it is determined whether both the character strings of the first character string and the second character string have been successfully extracted by the extraction processing unit **402** in **S503**. When it is determined that both the character strings have been successfully extracted, the process flow transitions to step **S505**. When the character strings have not been successfully extracted, the process flow in FIG. 5 ends.

[0061] In **S505**, the user information generation unit **403** generates the user information described in Table 1 by using the first character string and the second character string extracted in **S503**. Specifically, the user information generation unit **403** allocates the first character string to the value in the “user ID” column and the second character string to the value in the “password” column. The user information generation unit **403** also hashes the second character string to allocate an obtained value to the value in the “HashKey” column.

[0062] In **S506**, the user information generation unit **403** stores the user information described in Table 1 generated in **S505** in the secondary storage device **308**. The user information may be stored in a storage unit of the client terminal **101**. In **S507**, the input/output apparatus **310** displays the values in the “user ID” column and the “password” column of the user information generated in **S505**.

[0063] The registered user information is used for the user login to the image formation apparatus **102**, and the image formation apparatus **102** prints an image associated with a logged-in user.

[0064] As described above, based on the flowchart in FIG. 5, the user information can be directly generated based on the information described on the identification certificate. Thus, it is no longer necessary for an administrator of the image formation apparatus **102** to correct an input error of the user information. In addition, since the operation of the input/output apparatus **310** at the time of

the user information registration can be kept to a minimum, it becomes possible to efficiently register the user information.

[0065] As described above, according to the present embodiment, it is possible to reduce the operational burden on the administrator of the image formation apparatus **102** at the time of the user information registration.

Second Embodiment

[0066] According to the first embodiment, the information described on the identification certificate is extracted to be used as it is for the generation of the user information.

[0067] In the first embodiment, since even a person who has not been granted a use permission of the image formation apparatus **102** from the administrator of the image formation apparatus **102** can register the user information, additional instances of the user information may be registered in the image formation apparatus **102**. To limit users of the image formation apparatus **102**, the administrator of the image formation apparatus **102** needs to register the user information, which may lead to additional work on the part of the administrator in the registration of the user information.

[0068] According to a second embodiment, the administrator of the image formation apparatus **102** creates a job for activating the process flow described in FIG. 5, and a user who has been granted permission from the administrator can perform the registration of the user information by using the job.

[0069] FIG. 7 illustrates a software configuration according to the present embodiment. The external communication unit **404**, the printing unit **405**, the copying unit **406**, the boxing unit **407**, and the user information management unit **408** are equivalent to the functions described in FIG. 4 corresponding to the respective reference numerals, and thus the description thereof is not repeated herein.

[0070] A password generation unit **700**, a registration job generation unit **701**, and a registration job transmission unit **702** are functions included in the driver **400**. The password generation unit **700** generates a password for activating a job (hereinafter, a registration job) for executing the process flow illustrated in FIG. 5. There is no restriction on a format of a value of the password generated by the password generation unit **700**. The registration job generation unit **701** generates a registration job. The registration job will be described in detail with reference to FIG. 8.

[0071] FIG. 8 illustrates an example of a setting screen of the registration job that is displayed on the display **207**. Login device name **800** displays a name of the image formation apparatus **102** configured to execute the registration job. Device IP **801** displays an IP address of the image formation apparatus **102** configured to execute the registration job.

[0072] Registration job activation password **802** inputs the password for activating the registration job in the image formation apparatus **102**. Job transmission button **803** is a button for transmitting the registration job to the image formation apparatus **102** displayed under the login device name **800**. Job transmission button **803** can be selected in a case where the password is input to an input field of the registration job activation password **802**.

[0073] The description will now return to FIG. 7.

[0074] In a case where the password has been set in the registration job activation password **802** and the job transmission button **803** has been selected, the registration job transmission unit **702** transmits the registration job to the image formation apparatus **102** described under the login device name **800**. The registration job is transmitted to a registration job execution unit **703** via the external communication unit **404**.

[0075] Registration job execution unit **703** is a function for executing the registration job received from the registration job transmission unit **702**. Registration job execution unit **703** includes a registration job authentication unit **703-1**, a scanning unit **703-2**, an extraction processing unit **703-3**, and a user information generation unit **703-4**.

[0076] The registration job authentication unit **703-1** performs an authentication to activate the

registration job. In a case where content that has been input by the input/output apparatus **310** matches a password that has been set in the registration job activation password **802**, the registration job authentication unit **703-1** permits a use of the registration job. When the use of the registration job is permitted, a person who desires to use the image formation apparatus **102** scans the identification certificate by the scanning unit **703-2** to be allowed to perform the registration of the user information via the extraction processing unit **703-3** and the user information generation unit **703-4**.

[0077] The scanning unit **703-2**, the extraction processing unit **703-3**, and the user information generation unit **703-4** are respectively similar to the scanning unit **401**, the extraction processing unit **402**, and the user information generation unit **403**, and thus the description thereof is not repeated herein.

[0078] FIG. **9** illustrates a process sequence of the entire print system according to the present embodiment.

[0079] Steps **S901** to **S903** are started when the driver **400** has been activated in the client terminal **101**.

[0080] In **S901**, the password generation unit **700** generates a password to be attached to the registration job. In **S902**, the registration job generation unit **701** generates the registration job described in FIG. **8**. In **S903**, the registration job transmission unit **702** transmits the registration job generated in **S902** to the image formation apparatus **102** described under the login device name **800**.

[0081] Steps **S904** and **S905** are started when the image formation apparatus **102** has received the registration job from the client terminal **101**.

[0082] In **S904**, the registration job authentication unit **703-1** of the registration job execution unit **703** performs an authentication for activating the registration job. In a case where the value input from the input/output apparatus **310** matches the password set in the registration job, the registration job authentication unit **703-1** activates the registration job, and the process flow transitions to **S905**. In a case where the value is mismatched with the password, the process flow in FIG. **9** ends.

[0083] In **S905**, the registration job execution unit **703** executes the registration job to generate the user information from the information described on the identification certificate. In **S905**, steps **S500** to **S506** are executed.

[0084] In **S906**, the input/output apparatus **310** displays the user information generated in **S905**.

[0085] As described above, based on the sequence diagram in FIG. **9**, the user who has obtained the password from the administrator of the image formation apparatus **102** can perform the registration of the user information in the image formation apparatus **102** themselves. In other words, some of the user information registration operation can be delegated to the user who desires to use the image formation apparatus **102**. In addition, since only the user who has been granted the permission from the administrator of the image formation apparatus **102** registers the user information, it is possible to avoid registration of unnecessary user information.

[0086] As described above, according to the present embodiment, the operational burden on the administrator of the image formation apparatus **102** in the user registration operation can be reduced.

Third Embodiment

[0087] In the first embodiment, the user information is created based on the information extracted from the identification certificate to be registered in the image formation apparatus **102**.

[0088] In the first embodiment, additional instances of user information may be registered. For example, there can be a case where the registration of the user information has been performed, but thereafter, no login has been made to the image formation apparatus **102**. In such a case, the administrator of the image formation apparatus **102** performs an authority setting for the user information that is not to be used and delete the user information, which may result in additional

work on the part of the administrator of the image formation apparatus **102** increases.

[0089] According to a third embodiment, at the time of new registration of user information, the user information is put into a provisionally registered state, and when the user logs into the image formation apparatus **102** for the first time, an initial password of the user information is changed and then formal registration is performed. Based on this, according to the third embodiment, the input/output apparatus **310** is caused to display only the formally registered user information.

[0090] FIG. **10** illustrates a flowchart of provisional registration processing of the user information according to the present embodiment.

[0091] A realization method and a start condition for the flowchart in FIG. **10** is similar to those for the flowchart in FIG. **5**, and the description thereof is not repeated herein. Steps **S1000** to **S1004** are equivalent to steps **S500** to **S504**, and the description thereof is not repeated herein. In **S1005**, the user information generation unit **403** generates the user information. At this time, the user information generation unit **403** attaches a parameter indicating a registration state to the user information. An example of the user information generated in **S1005** is illustrated in Table 3.

TABLE-US-00003
TABLE 3 User information with status
User ID Password Status
Shimomaru
Taro 110100197810270022 Provisional

[0092] The “user ID” column is equivalent to the “user ID” column of the user information described in Table 1, and the description thereof is not repeated herein. The “password” column is a column for storing a password to be used at the time of login to the image formation apparatus **102**. In **S1005**, the national identification number **606** or the expiry **604** is stored in this column as the initial password.

[0093] A “status” column is a column for storing a value indicating a registration state of the user information. In a case where the value is “Provisional”, the user information is in the provisionally registered state. In a case where the value is “Active”, the user information is in a formally registered state. Only the user information with the value of “Active” is displayed on the input/output apparatus **310** and subjected to a management such as a set value alteration via the user information management unit **408**.

[0094] Steps **S1006** and **S1007** are equivalent to steps **S506** and **S507**, and the description thereof is similarly not repeated herein. In **S1006**, when the user logs into the image formation apparatus, the user information is registered.

[0095] FIG. **11** illustrates a flowchart of formal registration processing of the user information according to the present embodiment. Steps **S1100** to **S1106** are executed when the user information management unit **408** has received a signal associated with the login processing from the input/output apparatus **310**. In **S1100**, the user information management unit **408** obtains the user ID and the password for logging into the image formation apparatus **102** from the input/output apparatus **310**.

[0096] In **S1101**, the user information management unit **408** authenticates the login user. In a case where a combination of the user ID and the password received in **S1100** matches a combination of the values in the “user ID” column and the “password” column that are described in the user information with status in Table 3, it is determined that the authentication is successful, and the process flow transitions to step **S1102**. In a case where the combinations of the user ID and the password are not matched with each other, in other words, a case where the authentication has failed, the process flow in FIG. **11** ends.

[0097] In **S1102**, the status of the user information authenticated in **S1101** by the user information management unit **408** is checked. In a case where the value in the “status” column of the user information with status in Table 3 is “Provisional”, in other words, a case where the user information is in the provisionally registered state, the process flow transitions to step **S1103**. In a case where the value in the “status” column is “Active”, the process flow in FIG. **11** ends.

[0098] In **S1103**, the user information management unit **408** obtains a new password via the input/output apparatus **310**. In **S1104**, the user information management unit **408** determines

whether the password set in the user information with status is mismatched with the password obtained in **S1103**. When it is determined that the passwords are mismatched with each other, the process flow in **FIG. 11** transitions to step **S1105**. When it is determined that the passwords are matched with each other, the process flow returns to step **S1103**.

[0099] In **S1105**, the user information management unit **408** performs update processing of the user information with status. The value in the “password” column of the user information with status is updated to the password obtained in **S1103**. The value in the “status” column of the user information with status is updated to “Active”.

[0100] In **S1106**, the input/output apparatus **310** displays the user information on which the update processing has been performed in **S1105**. The input/output apparatus **310** displays the user ID of the user information and the new password updated in **S1105**.

[0101] As described above, based on the flowcharts in **FIG. 10** and **FIG. 11**, only the user who actually uses the image formation apparatus **102** can be subjected to the user information management by the administrator of the image formation apparatus **102**.

[0102] As described above, according to the present embodiment, the load of the user information management operation on the administrator of the image formation apparatus **102** can be reduced.

Fourth Embodiment

[0103] In the first embodiment, user information is newly created based on the information extracted from the identification certificate to be registered in the image formation apparatus **102**.

[0104] In the first embodiment, the information of the identification certificate of the user is updated to implement the registration of the user information by scanning the identification certificate again such that both the new user information and the old user information are registered in the image formation apparatus **102**. To delete the old user information, the administrator of the image formation apparatus **102** operates the image formation apparatus **102**. The administrator of the image formation apparatus **102** can change the setting of the registered user information via the user information management unit **408**. Updating the identification certificate may result in additional work on the part of the administrator of the image formation apparatus **102**.

[0105] According to a fourth embodiment, update of the user information is enabled in a case where the information described on the identification certificate is updated.

[0106] **FIG. 12** illustrates a flowchart of user information update processing of the image formation apparatus **102** according to the present embodiment. A realization method and a start condition for the flowchart in **FIG. 12** are similar to those for the flowchart in **FIG. 5**, and the description thereof is not repeated herein.

[0107] Steps **S1200** to **S1204** are equivalent to steps **S500** to **S504**, and the description thereof is not repeated herein.

[0108] In **S1205**, the user information generation unit **403** determines whether an owner of the identification certificate scanned in **S1200** is registered in the image formation apparatus **102** as the user. Specifically, the user information generation unit **403** accesses the user table of Table 2 stored by the user information management unit **408**. Thereafter, the user information generation unit **403** searches the “HashKey” column in the user table to attempt to obtain the user information in Table 1 having the same value as the hash value of the national identification number **606** extracted in **S1203**. In a case where the user information has been successfully obtained, the process flow transitions to step **S1206**. In a case where the user information has not been successfully obtained, the process flow transitions to step **S1207**.

[0109] In **S1206**, the user information generation unit **403** implements an update of the user information in Table 1. Specifically, the user information generation unit **403** replaces the value in the “user ID” column of the user information in Table 1 with the character string relevant to the name **600** obtained in **S1203**.

[0110] Steps **S1207** and **S1208** are equivalent to steps **S505** and **S506**, and the description thereof is not repeated herein.

[0111] In **S1209**, the input/output apparatus **310** displays the values in the “user ID” column and the “password” column of the user information. Specifically, in a case where the process flow transitions from **S1206** to **S1209**, the input/output apparatus **310** displays the new name **600** and the unchanged password. In a case where the process flow transitions from **S1208** to **S1209**, the input/output apparatus **310** displays the name **600** and the national identification number **606** or the expiry **604**.

[0112] As described above, based on the flowchart in FIG. **12**, the update processing of the user information can be implemented by scanning the identification certificate.

[0113] As described above, according to the present embodiment, it is possible to reduce the operational burden on the administrator of the image formation apparatus **102** at the time of the update of the information of the identification certificate.

Fifth Embodiment

[0114] In the second embodiment, by introducing the job for activating the process flow described in FIG. **5**, the registration operation of the user information is performed by the person who desires to use the image formation apparatus **102**.

[0115] In the second embodiment, a plurality of image formation apparatuses **102** are configured to register user information, which results in the administrator of the image formation apparatuses **102** issuing the same number of registration jobs of FIG. **8** as the number of image formation apparatuses **102**.

[0116] User information of the same user may differ between the plurality of image formation apparatuses **102**.

[0117] According to a fifth embodiment, the registration job is transmitted to a plurality of image formation apparatuses **102**, and when the registration job is executed in one of the image formation apparatuses **102**, the user information is to be shared with the other image formation apparatus **102**.

[0118] FIG. **13** illustrates an example of the setting screen of the registration job that is displayed on the display **207** according to the present embodiment.

[0119] Login device name **1300** is a check box for displaying identification information of one or a plurality of image formation apparatuses **102** that are set as targets to which the registration job is to be transmitted. Thus, the registration job can be transmitted to the plurality of image formation apparatuses **102** at the same time.

[0120] Registration job activation password **1301** and job transmission button **1302** are equivalent to the registration job activation password **802** and the job transmission button **803** in FIG. **8**, and the description thereof is not repeated herein.

[0121] FIG. **14** illustrates an entire sequence according to the present embodiment. Step **S1401** is equivalent to step **S901**, and the description thereof is not repeated herein. In **S1402**, the registration job generation unit **701** generates the registration job illustrated in FIG. **13**. In **S1403**, the registration job transmission unit **702** transmits the registration job of FIG. **13** to all the image formation apparatuses **102** selected under the login device name **1300**.

[0122] Steps **S1404** to **S1406** are executed by an image formation apparatus **102-1**. The image formation apparatus **102-1** refers to a single image formation apparatus **102** that received the registration job in **S1403** and in which the registration job has been activated by the registration job execution unit **703**.

[0123] Step **S1404** is equivalent to step **S904**, and the description thereof is not repeated herein. In **S1405**, the registration job execution unit **703** newly registers or updates the user information by executing the process flow described in FIG. **12**. In **S1406**, the input/output apparatus **310** displays the user information newly registered or updated in **S1405**.

[0124] In **S1407**, the user information management unit **408** of the image formation apparatus **102-1** transmits the user information registered in **S1405** to one or a plurality of image formation apparatuses **102-2** via the external communication unit **404**. Herein, the image formation apparatus **102-2** refers to one or a plurality of image formation apparatuses **102** that have been selected under

the login device name **1300** and have received the registration job, but in which the registration job has not been activated.

[0125] In **S1408**, the image formation apparatus **102-2** stores the user information received from the image formation apparatus **102-1** in the secondary storage device **308**.

[0126] FIG. **15** illustrates a flowchart at a time when the image formation apparatus **102-2** receives user information from the image formation apparatus **102-1** to register the user information according to the fifth embodiment.

[0127] In **S1500**, the user information management unit **408** of the image formation apparatus **102-2** refers to the secondary storage device **308** to obtain the user table of Table 2.

[0128] In **S1501**, the user information management unit **408** of the image formation apparatus **102-2** determines whether the value in the “HashKey” column of the user information received in **S1406** is included in the “HashKey” column in the user table. When it is determined that the value is included, the flowchart transitions to step **S1502**. When it is determined that the value is not included, the flowchart transitions to step **S1503**.

[0129] In **S1502**, the user information management unit **408** of the image formation apparatus **102-2** updates the user information to store the updated user information in the secondary storage device **308**. Specifically, the user information management unit **408** of the image formation apparatus **102-2** replaces the value in the “user ID” column of the registered user information with the value in the “user ID” column of the received user information.

[0130] In **S1503**, the user information management unit **408** of the image formation apparatus **102-2** adds user information to store the user information in the secondary storage device **308**.

Specifically, the user information management unit **408** of the image formation apparatus **102-2** stores the user information received in **S1406** in the secondary storage device **308** as is.

[0131] As described above, based on the registration job in FIG. **13**, the sequence diagram in FIG. **14**, and the flowchart in FIG. **15**, it is possible to register the user information for the plurality of image formation apparatuses **102**. The content of the user information of the same person can be synchronized between the plurality of image formation apparatuses **102**.

[0132] As described above, according to the present embodiment, in a case where the plurality of image formation apparatuses **102** are present, the user registration operational burden on the administrator of the image formation apparatuses **102** can be reduced. In addition, it is possible to remove the risk of confusing the administrator of the image formation apparatuses **102** because of the difference in the content of the user information of the same person between the plurality of image formation apparatuses **102**.

Sixth Embodiment

[0133] The second embodiment introduces the job for activating the process flow described in FIG. **5**, and the registration operation of the user information is performed by the person who desires to use the image formation apparatus **102**.

[0134] In the second embodiment, the administrator of the image formation apparatus **102** performs the operation of granting the authority for using the function of the image formation apparatus **102** to each user information the same number of times as the number of users who are registered in the image formation apparatus **102**, which may result in an increase in the work of the administrator.

[0135] According to the present embodiment, the authority for using the function of the image formation apparatus **102** is granted to the user information based on the registration job activation password **802** set in the registration job.

[0136] FIG. **16** illustrates a flowchart in the user information registration processing of the present embodiment.

[0137] A realization method and a start condition for the flowchart in FIG. **16** are similar to those for the flowchart in FIG. **5**, and the description thereof is not repeated herein. Steps **S1600** to **S1604** are equivalent to steps **S500** to **S504**, and the description thereof is not repeated herein.

[0138] In **S1605**, the user information generation unit **703-4** obtains, from the user information

management unit **408**, an authority map in which a necessary authority to use the function of the image formation apparatus **102** corresponding to the registration job activation password **802** is stored. An example of the authority map is illustrated in Table 4.

TABLE-US-00004 TABLE 4 Authority map Authority setting Password Printing: permitted, copying: permitted, boxing: 1111 permitted, scanning: permitted Printing: not permitted, copying: not permitted, 0000 boxing: not permitted, scanning: not permitted

[0139] An “authority setting” column is a column for storing a necessary authority to use the function of the image formation apparatus **102** as a built-in set value.

[0140] “Password” column is a column for storing a password to be set as the registration job activation password **802**. In **S1606**, the user information generation unit **703-4** determines whether the password that has been set as the registration job activation password **802** is present in the “password” column of the authority map in Table 4. When it is determined that the password is present, the flowchart transitions to step **S1607**. When it is determined that the password is not present, the flowchart transitions to step **S1608**.

[0141] In **S1607**, the user information generation unit **703-4** obtains an authority setting corresponding to the registration job activation password **802** from the authority map in Table 4. For example, in a case where “1111” is set as the registration job activation password **802**, the user information generation unit **703-4** obtains an authority setting in which the use of all of printing, copying, boxing, and scanning is permitted.

[0142] In **S1608**, the user information generation unit **703-4** generates user information with an authority setting by using the first character string and the second character string extracted in **S1503**, and also the authority setting obtained in **S1607**. An example of the user information with authority setting is illustrated in Table 5. “1111” is set as the registration job activation password. TABLE-US-00005 TABLE 5 User information with authority setting User ID Password Authority setting Shimomaruko 110100197810270022 Printing: permitted, Taro copying: permitted, boxing: permitted, scanning: permitted

[0143] The “user ID” column and the “password” column are equivalent to the “user ID” column and the “password” column of the user information in Table 1, and the description thereof is not repeated herein.

[0144] The “authority setting” column is a column for storing the authority setting obtained in **S1607**. In a case where the process flow directly transitions from **S1606** to **S1608**, a value in the “authority setting” column is empty.

[0145] In **S1609**, the user information generation unit **703-4** stores, in the secondary storage device **308**, the user information with authority setting in Table 5 that is generated in **S1608**. In **S1610**, the input/output apparatus **310** displays all the columns of the user information with authority setting in Table 5 that is generated in **S1608** and the value in each of the columns.

[0146] As described above, based on the flowchart in FIG. 15, the user information granted with the authority to use the function of the image formation apparatus **102** can be generated.

[0147] As described above, according to the present embodiment, it is possible to reduce the operational burden on the administrator of the image formation apparatus **102** when the authority is granted to the user information.

OTHER EMBODIMENTS

[0148] The present disclosure can also be realized by processing in which a program for realizing one or more functions of the above-described embodiments is supplied to a system or an apparatus via a network or a storage medium, and one or more processors in a computer of the system or the apparatus read out and execute the program. The present disclosure can be realized by a circuit configured to realize one or more function (for example, an application specific integrated circuit (ASIC)).

[0149] The disclosure of the above-described embodiments includes the following configurations.

[0150] Configuration 1. An image formation apparatus including: [0151] a reading unit configured

to read a sheet on which a character string is printed; and [0152] a memory storing a program and a processor, that when executing the program, is configured to extract the character string from the sheet read by the reading unit, to generate, based on the character string extracted by the extraction unit, login information for logging into the image formation apparatus; and to register the login information generated by the generation unit.

[0153] Configuration 2. The image formation apparatus according to Configuration 1, the processor further configured to log into the image formation apparatus based on the registered login information.

[0154] Configuration 3. The image formation apparatus according to Configuration 2, further including a printing unit configured to print an image associated with a user who has logged in.

[0155] Configuration 4. The image formation apparatus according to Configuration 1, in which the character string is personal information.

[0156] Configuration 5. The image formation apparatus according to Configuration 1, in which the sheet is an identification certificate.

[0157] Configuration 6. The image formation apparatus according to Configuration 1, the processor further configured to update, in a state in which the login information has been registered, in a case where the login information is generated, the registered login information to the generated login information.

[0158] Configuration 7. The image formation apparatus according to Configuration 1, in which the login information includes user information and a password.

[0159] Configuration 8. A method for an image formation apparatus, the method including: [0160] reading a sheet on which a character string is printed; [0161] extracting the character string from the read sheet; [0162] generating, based on the extracted character string, login information for logging into the image formation apparatus; and [0163] registering the generated login information.

[0164] Configuration 9. The method according to Configuration 8, further including logging into the image formation apparatus based on the registered login information.

[0165] Configuration 10. The method according to Configuration 9, further including printing an image associated with a logged in user.

[0166] Configuration 11. The method according to Configuration 8, in which the character string is personal information.

[0167] Configuration 12. The method according to Configuration 8, in which the sheet is an identification certificate.

[0168] Configuration 13. The method according to Configuration 8, further including updating, in a state in which the login information has been registered, in a case where the login information is generated, the registered login information to the generated login information.

[0169] Configuration 14. The method according to Configuration 8, in which the login information includes user information and a password.

[0170] Configuration 15. A non-transitory computer-readable recording medium recording a program for causing a computer to execute a method for an image formation apparatus, the method including: [0171] reading a sheet on which a character string is printed; [0172] extracting the character string from the read sheet; [0173] generating, based on the extracted character string, login information for logging into the image formation apparatus; and [0174] registering the generated login information.

[0175] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for

example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0176] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0177] This application claims the benefit of Japanese Patent Application No. 2024-035727, filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image formation apparatus comprising: a reading unit configured to read a sheet on which a character string is printed; and a memory storing a program and a processor, when executing the program, configured to: extract the character string from the read sheet, generate, based on the extracted character string, login information for logging into the image formation apparatus, and register the generated login information.
2. The image formation apparatus according to claim 1, wherein the processor is further configured to enable a login into the image formation apparatus based on the registered login information.
3. The image formation apparatus according to claim 2, further comprising a printing unit configured to print an image associated with a logged in user.
4. The image formation apparatus according to claim 1, wherein the character string is personal information.
5. The image formation apparatus according to claim 1, wherein the sheet is an identification certificate.
6. The image formation apparatus according to claim 1, wherein the processor is further configured to update, in a state in which the login information has been registered, in a case where the login information is generated, the registered login information to the generated login information.
7. The image formation apparatus according to claim 1, wherein the login information includes user information and a password.
8. A method for an image formation apparatus, the method comprising: reading a sheet on which a character string is printed; extracting the character string from the read sheet; generating, based on the extracted character string, login information for logging into the image formation apparatus; and registering the generated login information.
9. The method according to claim 8, further comprising logging into the image formation apparatus based on the registered login information.
10. The method according to claim 9, further comprising printing an image associated with a logged in user.
11. The method according to claim 8, wherein the character string is personal information.
12. The method according to claim 8, wherein the sheet is an identification certificate.
13. The method according to claim 8, further comprising updating, in a state in which the login information has been registered, in a case where the login information is generated, the registered

login information to the generated login information.

14. The method according to claim 8, wherein the login information includes user information and a password.

15. A non-transitory computer-readable recording medium recording a program for causing a computer to execute a method for an image formation apparatus, the method comprising: reading a sheet on which a character string is printed; extracting the character string from the read sheet; generating, based on the extracted character string, login information for logging into the image formation apparatus; and registering the generated login information.
